

RAHUL MITRA

ENGINEER - SOFTWARE, AI/ML, OPERATIONS RESEARCH & SECURITY

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PROFILE SUMMARY

Early-career engineer and current NUS undergraduate (BE Industrial Systems Engineering, 2nd Major Computer Science, Minor Mathematics; graduating Jul 2027) working across the full technical stack: applied AI/ML (RAG, AI agents, fine-tuned transformers, HPC), solution architecture and cloud, operations research and optimization, and adversarial security. Proven through hands-on delivery: led the fine-tuning of a 109M-parameter BERT model on the ASPIRE 2A supercomputer, maintains the AsyncDDGS PyPI library, and shipped a Singpass/Myinfo-integrated civic balloting platform now production-ready for national rollout and winner of the S\$20,000 Sparks Community Innovation Fund. Also an active bug bounty researcher on YesWeHack against Singapore government and transport targets.

EDUCATION

National University of Singapore	Aug 2023 - Jul 2027
- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics - Ranked 1st of 24 students in 3D Computer Vision (AY2025/26 Semester 2), earning the NUS School of Computing Certificate of Outstanding Performance. - Relevant coursework: Software Engineering, Artificial Intelligence, 3D Computer Vision, Data Structures and Algorithms, Simulation, Statistics, Operations Research, Linear Algebra, Probability, Computer Systems, Mathematics, Engineering Systems, Industrial Systems Engineering.	
Anderson Serangoon Junior College (ASRJC)	Feb 2019 - Dec 2020
- Singapore-Cambridge GCE A-Level	

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd	Jun 2026 - Present
- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation. - Strengthen monitoring and process structure across healthcare supply chains with AI, helping ensure reliable product availability for vulnerable patient groups.	
Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd	Jan 2026 - Jun 2026
- Built a Python digital-twin simulation and constraint-programming optimizer for an NP-hard hybrid flow-shop scheduling problem, translating production complexity into decision-ready schedules. - Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.	
Software Engineer (Citizen Developer), People's Association	Sep 2025 - Present
- Awarded the S\$20,000 Sparks Community Innovation Fund for building an end-to-end digital garden-plot balloting platform (project codename Churp) spanning admin and public-facing frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore. - Integrated Singpass Login (OIDC authorization-code flow / Relying Party) and Myinfo (v5) for passwordless authentication and consent-based retrieval of verified identity (NRIC/FIN) and registered address, automating resident eligibility checks. - Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders.	
Independent Bug Bounty Researcher, YesWeHack Platform	May 2024 - Present
- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses. - Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports using Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts. - Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.	
Base Support Assistant, Republic of Singapore Navy	Jan 2022 - Jan 2023
- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment. - Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.	
SELECTED PROJECTS AND COMPETITIONS	
Maritime Deficiency Severity Forecasting - Maritime Hackathon 2025 (Team Lead) - Hackathon	2025
- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with additional DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer to forecast inspection deficiency severity.	
Volt Pulse SG - SMU Hack For Cities 2026 (Top 8 finalist of 50+ teams) - Hackathon	Jan 2026
- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.	
AsyncDDGS - Open-Source Search Client - PyPI library	2024 - Present
- Authored and maintain an asyncio-first, aiohttp-based DuckDuckGo client published on PyPI with pytest CI/CD, adopted across AI tooling, Discord bots, and search projects.	
AgeWellLah.AI - NUS HealthHack 2025 (AI/ML Engineer) - Hackathon	2025

- Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.

SmartExam - GovTech x NTUPC Product Hackathon - Hackathon

2025

- Built a multi-agent autonomous exam-generation system with a RAG pipeline and Next.js frontend, automating source ingestion, question generation, and education workflows.

EthosLens - LifeHack 2025 (Backend Developer) - Hackathon

2025

- Built an AI research engine scoring brand ethics and sustainability using FastAPI, Python, and LangChain with automated evidence retrieval.

DSTA BRINHACK 2025 CTF (top quartile of 400+ teams) - Capture-the-flag competition (first CTF)

2025

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, finishing in the top quartile among 400+ teams.

OnTheSpectrum - Codex & Blender MCP 3D Pipeline - Personal

May 2026

- Built a local-first Blender-to-Three.js asset pipeline using Codex and Blender MCP, generating GLB assets, metadata, previews, playable-world QA, and Remotion video workflows.

IE2110 Operations Research Graph Optimization - Coursework

2024

- Built LP, network-flow, and nonlinear programming models, including a Jupyter SSSP/APSP (Dijkstra) graph optimization for decision support.

Arcane Security Tooling Family - Personal

Mar 2026

- Built Rust-forward security and red-team tooling (Arcane, Arcane-PP, Arcane-GLM, Arcane-OCR) for automation, OCR, and adversarial workflows.

Additional projects - Personal and coursework

2023 - 2026

- Project Utopia - real-time global situational-awareness dashboard with AI news aggregation, geopolitical monitoring, and geospatial visualization (Mar 2026).
- Interactive Portfolio Infrastructure - AI-assisted portfolio in React, TypeScript, Three.js, and Matter.js with profile switching, searchable field notes, and analytics (2026).
- NVIDIA Disaster Risk Monitoring with Satellite Imagery - geospatial AI with U-Net semantic segmentation on GPU/ASPIRE 2A workflows (2023).
- Fine-tuning LLMs for Cybersecurity - coursework exploring Mistral/LLaMA fine-tuning with AutoTrain and AutoGen (2024).

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, Bash, C++, C#, Excel VBA, SQL

AI/ML: BERT, DNN, Transformers, RAG / RAG Architecture, AI Agents, OpenAI GPT-4, Gemini, SEALION, LangChain, Vector Search, Computer Vision, Model Evaluation, NLP, Deep Reinforcement Learning (PPO, A2C, DDPG, DQN, Double DQN), ASPIRE 2A / HPC / CUDA / distributed training

Operations Research / Optimization: Constraint Programming, Scheduling Optimization, Hybrid Flow Shop, Digital Twin Simulation, Simulation, Dijkstra / SSSP / APSP, Decision Analytics, Statistics, Jupyter, Mathematical Modeling (Linear Algebra, Probability, Calculus)

Cloud / DevOps: Azure, Docker, CI/CD, Bitbucket CI/CD, GitHub Actions, Supabase, API Design, Codex

Security: Burp Suite, Wireshark, SSRF, CSRF, SQL Injection, NoSQL Injection, Authentication Bypass, OAuth2/JWT, OWASP Top 10, Secure Design, Fuzzing, Reverse Engineering, Custom Exploit Scripting, Cloud Security (AWS IMDSv2, Metadata APIs)

Web / Full-Stack: React, Next.js, Three.js, Matter.js, FastAPI, Flask, aiohttp, asyncio, GIS Mapping, Singpass Login & Myinfo (OIDC), Konva.js

Solution Architecture / Product: Solution Architecture, Manufacturing Systems, Stakeholder Discovery, Rapid Prototyping, Accessibility, Product Discovery, User Testing, Dashboards

Hardware / Embedded: Edge AI / Hailo model deployment, Autonomous robotics (sensor integration)

- Disaster Risk Monitoring Using Satellite Imagery - NVIDIA Deep Learning Institute (Dec 2023)
- Deep Reinforcement Learning Hands-On (DQN, Double DQN, DDPG, PPO, A2C) - Packt (2023)
- Docker for the Absolute Beginner - Hands-On - Stone River eLearning (2024)
- Reverse Engineering Windows Executables (digital forensics) (2024)
- Excel 2019 Advanced - Intellectzy (2023)
- Deep Learning CNN with Python - Packt (2023)
- Deep Learning RNN with Python - Packt (2023)
- Learning C# by Developing Games with Unity - Packt (2023)

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Support senior engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

Mar 2019 - Aug 2020

- Led astronomy outreach and organized weekly stargazing and observation programs.
- Competed in the 7th Singapore Astronomical Olympiad and engineered an autonomous robot for the National Robotics Challenge.
- Received the ASRJC Outstanding Contribution Award.